

PALCapital

Turning Sovereign Orders into Instant Working Capital

Sovereign Maritime Tokenization Engine

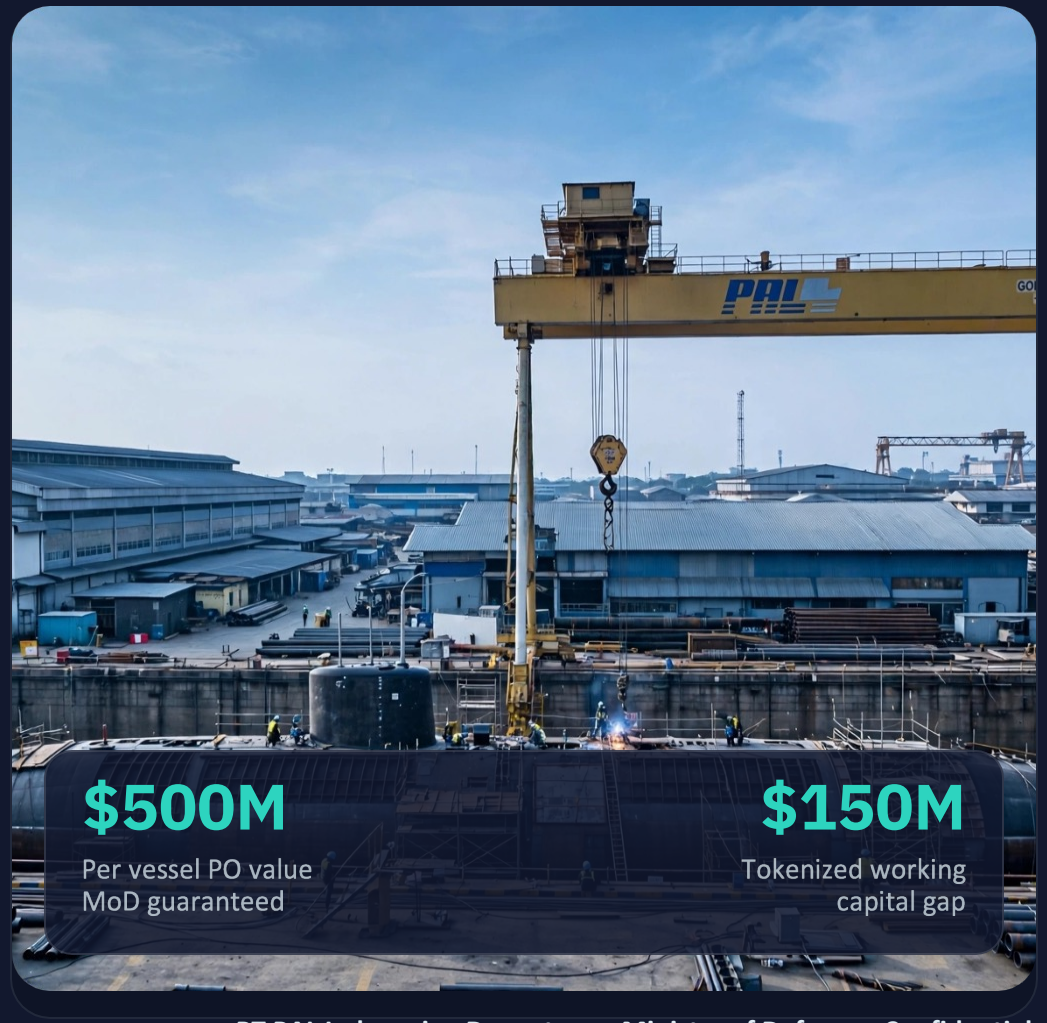
PT PAL Indonesia - Danantara – Ministry of Defense - Confidential

PT PAL holds $ Billions USD in Guaranteed Contracts — yet faces a Working Capital Gap

- **Sovereign Backlog:** 2x Scorpène Evolved ($2.16B) + 2x Merah Putih Frigate AH-140 ($1.2B) + LPDs & Hospital Ships - all MoD-guaranteed, 60-96 month cycles
- **Upfront Cash Outlay**: HY-80 steel, LIB arrays, propulsion & combat systems must be paid before milestone payments => creates $150-300M gap per program
- **75% materials imported** — IDR Trilliuns industry capex required. Forex and supply-chain risk concentrated.
- **Bank syndicates**: 6-9 months, heavy covenants, **9-11% all-in cost**. Too slow for sovereign build tempo.

96M
Build months per sub — cash negative for 24+ months

7-14D
PALCapital funding target vs 270 days bank



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Indonesia's $900B sovereign engine is looking for sovereign assets

$900B+ AUM

Danantara Indonesia launched Feb 24, 2025 by President Prabowo

1,000+ SOEs

Under management incl. PT PAL — 100% SOE

$20B Initial

Deployment mandate for strategic industries

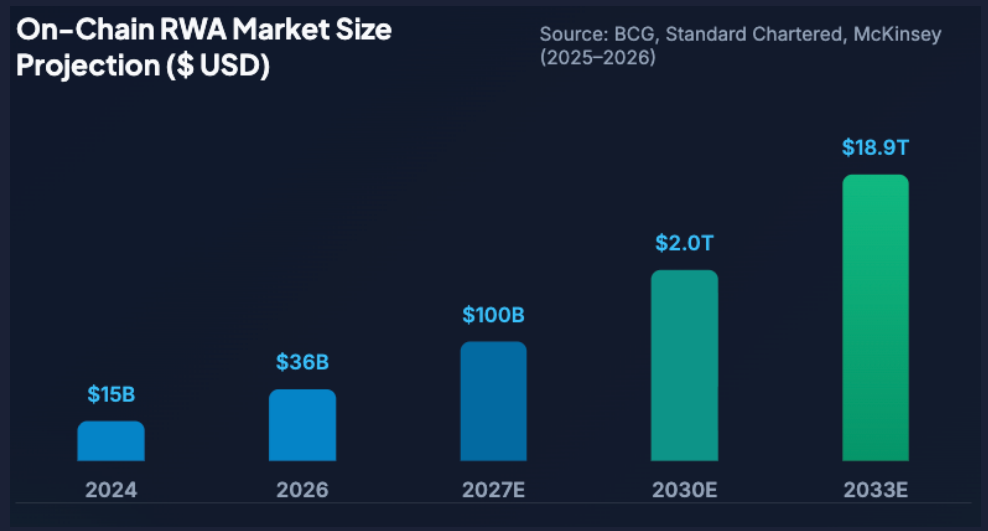
5 Pillars

Renewable, advanced mfg, downstream, food, AI & defense

- **PALCapital is first instrument to directly channel Danantara liquidity into defense shipbuilding with yield** — not grant, not debt, but tokenized receivable.
- **Not a** grant, not debt, A tokenized receivable backed by verified MoD purchase orders.
- Anchor investor: Danantara. Co-investors: Mandiri, BRI, sovereign funds. 100% state-aligned.

On-Chain Real World Asset Market Size Projections

$36B today → $18.9T by 2033. PT PAL can Lead Defense



Include tokenized private credit, sovereign bonds, trade finance, and gov procurement receivables



$24B
Jun 2025 Actual
Fastest segment: sovereign debt

$36B
2026E
Institutional inflection

$18.9T
2033E
BCG x Ripple Forcast

Standard Chartered	**$2T by 2028**
BCG x Ripple	**$18.9T by 2033**
Long-term	**$30T by 2034**

Live now

BlackRock · Franklin Templeton · Apollo

Why now:

- ✓ Tokenized sovereign debt is fastest-growing RWA segment
- ✓ Institutional adoption BlackRock, Franklin Templeton now live.
- ✓ Standard Chartered projects $2T by 2028, BCG x Ripple $18.9T by 2033.
- ✓ Indonesia can lead defense vertical - first mover advantage

Why Legacy Finance Fails Shipyards

Bank syndicates were designed for real estate, not 96-month sovereign submarine builds with classified workstreams.

Traditional Bank Syndicate

6-9 MONTHS

SPEED
180–270 days credit committee, covenants, collateral appraisal

COST
9–11% all-in + 2-3% fees, cash collateral lock

TRANSPARENCY
Opaque use-of-funds quarterly manual audit

CONTROL
Covenants + equity pledge risk state dilution risk

COMPLIANCE
Manual Edit

PALCapital STO

7–14 DAYS

SPEED
7–14 days KYB + ONCHAINID mint, instant settlement

COST
6–7% fixed yield no collateral, no syndication fee

TRANSPARENCY
Direct-to-vendor escrow (Mandiri Custody) + BKI 2-of-2 multi-sig + IoT evidence

CONTROL
100% state ownership retained, no withdrawToPAL() function

COMPLIANCE
Real-time on-chain audit for OJK/MoD + immutable trail for IKD sandbox

Result: Saves $8–12M per vessel in fees/interest, unlocks yard throughput +40% via automated supplier payments. Chart: *Time to funding 270d → 10d, cost 10.5% → 7%, transparency 2/10 → 10/10.*

Not crypto. Regulated Institutional Security Token

PERMISSIONED STO • ERC-3643 T-REX • HYPERLEDGER BESU PRIVATE L2

- Permissioned STO on Hyperledger Besu private L2 - 4 validators (PAL x2, Danantara, BKI) need 3 to finalize, 500 TPS, 99.95%
- ERC-3643 T-REX standard - SEC-acknowledged, compliance enforced at token level (auto-revert non-whitelisted)
- Asset: PAL-SUB-1 - yield-bearing debt note backed by verified MoD PO via ZK-SNARK (prove $500M value without exposing sonar/hull)
- Custody: WalletConnect v2, MetaMask Institutional, Fireblocks MPC - institutional grade



TOKEN TICKER	PAL-SUB-1 • 700 bps • 720 days
LEGAL STRUCTURE	Debt note • MoD PO collateral • Burn on settlement
PRIVACY LAYER	ZK verified • Classified specs never on-chain

WalletConnect v2 | MetaMask Inst. | Fireblocks

How it works — 6-step lifecycle from order to settlement



1

Trigger: PO Upload

MoD contract (including terms of payment) ingested, hashed on-chain. Trigger for tokenization.

2

ZK Shield

ZK-SNARK proves value & MoF signature. Zero classified specs leak.

3

Onboard

KYB/AML Verihubs, ONCHAINID (ERC734/735) minted, ERC-3643 whitelist - non-whitelisted auto-revert

4

Escrow Funding

$150M locked in smart escrow (on-chain, no owner, HSM) + fiat escrow Bank Mandiri Custody under Tripartite PAL-Danantara-Mandiri - determined by Danantara IC + MoF + OJK, not PAL

5

Build & Release

BKI Lead Surveyor + TNI AL Officer 2-of-2 multi-sig with Secure Enclave key + geotagged 4K photo + GPS + timestamp. IoT (crane load 150T, CNC cut %) as supporting evidence only in Phase 1 - weld quality remains BKI manual. Funds DIRECT to Krakatau Steel / vendor, bypass PAL

6

Settlement

Each MoD termin payment goes to Mandiri Collection Account first (hedging), then waterfall: pay accrued yield to investors first, forward remainder to PAL Treasury. At final delivery: principal + final yield + burn token.

Permissioned Besu L2 Privacy by design Chainlink + ISO20022 oracle

Investor Safety - PAL Never Gets a Blank Check, No Backdoor

Funds locked in smart escrow. Tranches release only after physical proof — direct to vendor, bypassing operational accounts.

T1 Steel Cutting — $40M

BKI + TNI AL 2-of-2 multi-sig (Secure Enclave key, geotagged photo, GPS) -> DIRECT to Krakatau Steel BRI (bypass PAL) - No withdrawToPAL() function exists

Geotagged evidence → Krakatau Steel

T2 Keel Laying — $45M

BKI + Navy 2-of-2 + crane load 150T as evidence (IoT as evidence only, not auto-trigger) -> engine vendor

Oracle auto-release → engine procurement

T3 Launch — $40M

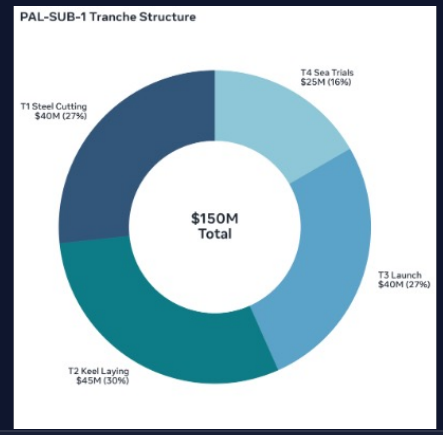
BKI + Navy + SAP S/4HANA status Launched -> systems integration vendor

Systems integration vendor

T4 Sea Trials — $25M

Final sign-off BKI + Navy -> outfitting vendor Guarantee: All MoD payments go to Mandiri Escrow first, funds bypass PAL operational accounts, zero diversion

Outfitting & commissioning





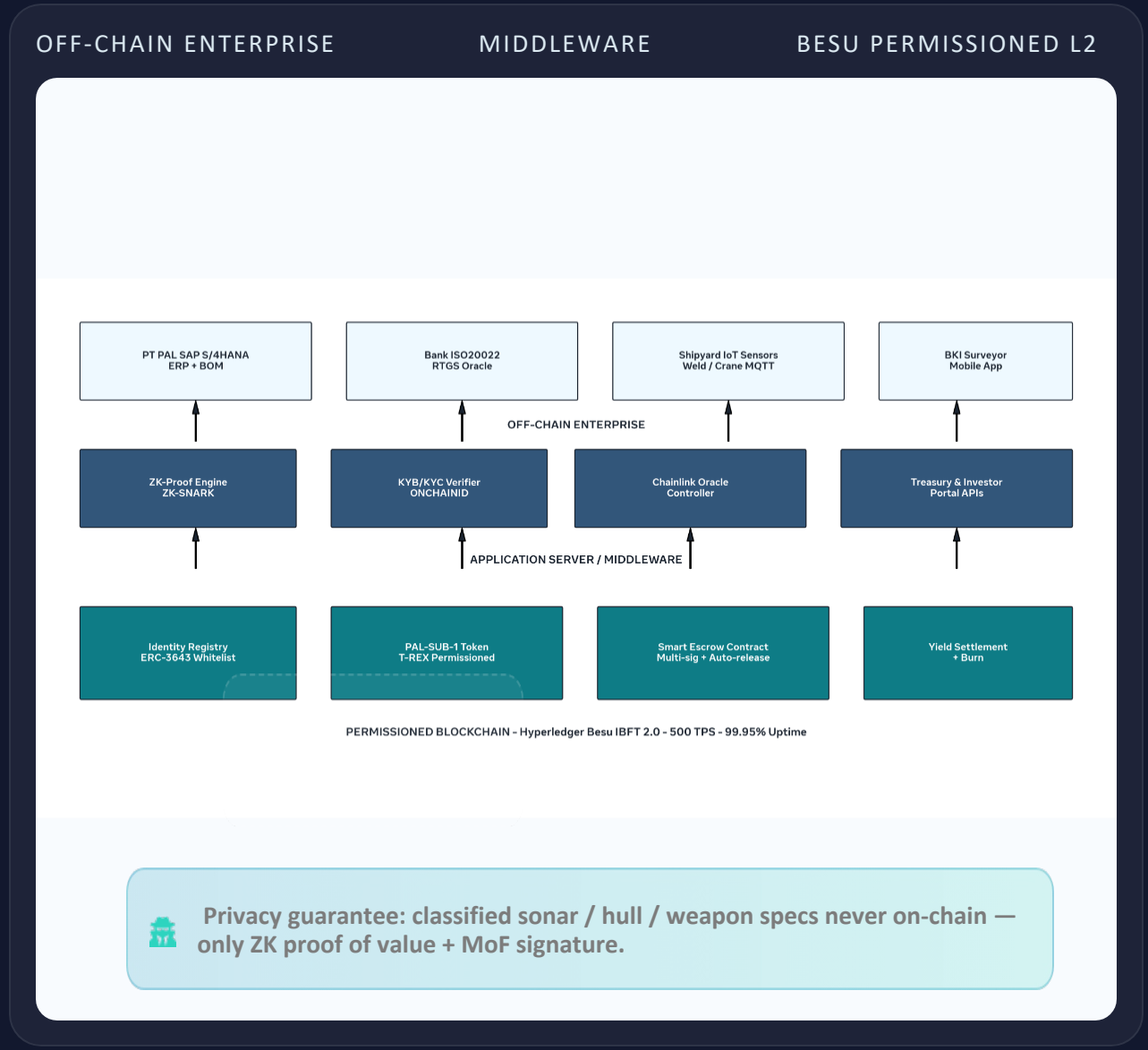
BKI SURVEYOR ORACLE

Geotagged + biometric signed proof before any release

- GPS + timestamp watermark mandatory
- 2FA hardware key, Secure Enclave
- Zero diversion — erc-3643 audit trail

Guarantee: $150M never touches PAL main account. Smart contract routes directly to suppliers. 3 Locks: Code (no withdraw), Validator (need 3 of 4), Upgrade (3-day timelock + 3-of-5 multisig visible to OJK)

Security & Compliance Architecture



Off-chain enterprise

SAP S/4HANA ERP + BOM, Bank ISO20022 RTGS oracle, Shipyard IoT MQTT (weld / crane), BKI Mobile App geotagged evidence

Middleware — application server

ZK-SNARK proof engine, KYB/KYC + ONCHAINID, Chainlink oracle controller, Treasury & Investor Portal APIs — K8s on BSSN-compliant sovereign cloud

On-chain — Hyperledger Besu IBFT 2.0

4 validators, 500 TPS, 99.95% uptime, Identity Registry ERC-3643 whitelist, PAL-SUB-1 Token T-REX permissioned, Smart Escrow multi-sig + auto-release, Yield Settlement + Burn

Security

AES-256-GCM at rest, TLS 1.3, mTLS, FIPS 140-2 L3 HSM Thales Luna, Slither/Mythril 100% coverage, BSSN + ISO 27001, Proxy ERC-1967 with 3-day timelock + 3-of-5 multisig DAO (PAL+Danantara+BSSN)

Trust

Smart Escrow on-chain = autonomous no owner, Fiat Escrow off-chain = Bank Mandiri Custodian owned under Tripartite Deed PAL-Danantara-Mandiri, determined by Danantara IC + MoF + OJK. 4 validators (PAL x2, Danantara, BKI) need 3 to finalize - PAL alone cannot move chain. Auditor nodes read-only MoD + OJK Blockscout

Financials & Yield Waterfall

PAL-SUB-1 • 700 BPS • 720 D

PRO-RATA YIELD CALCULATION — ON-CHAIN

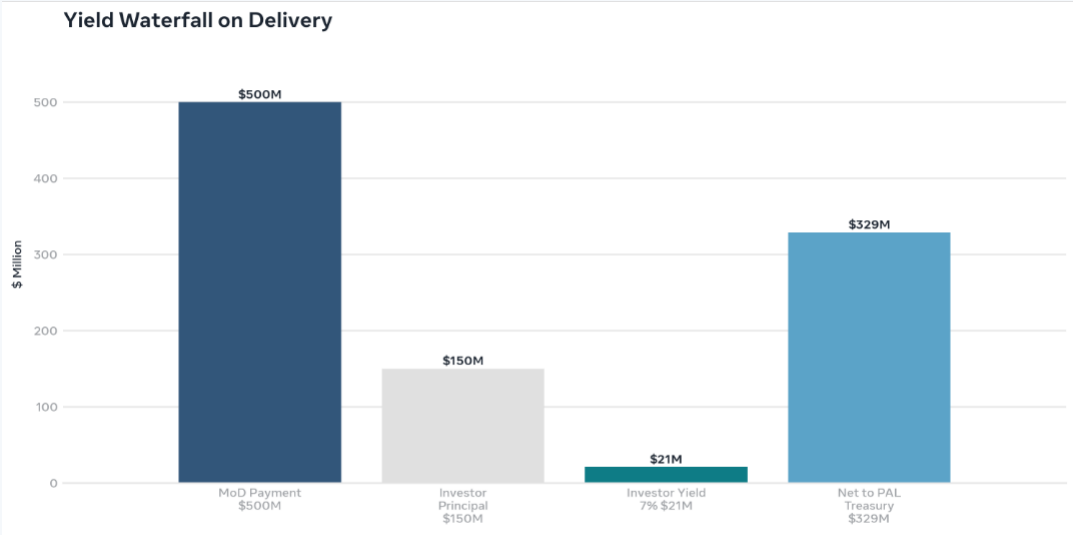
Total Return = Principal + (Principal × YieldBps/10000 × Duration/365)
$150M × 700bps × 720/365 = $21M yield

$150M → $21M
Principal locked 2 years at 7% p.a. Fixed, no floating

+200 bps
Early completion bonus if delivery >90 days early

Collection Account Model (Hedging for Investor): All MoD termin payments (DP 15% + T1-T5) go to Mandiri Escrow Collection Account, not PAL. Smart Contract Waterfall per termin:

1. **Pay accrued yield pro-rata to investors first (e.g., T1 $40M in -> $5.17M yield paid)**
2. **Forward remainder to PAL Treasury for ops (e.g., $34.83M to PAL)**
3. **At Final Delivery: Pay remaining Principal $150M + final yield, Burn PAL-SUB-1, Forward final remainder ~$329M to PAL. Investor is secured creditor - even if PAL defaults, future MoD payments still pay investor first**



Saves $8-12M / vessel
vs bank 9-11% + fees + collateral

$900B anchor
Danantara + sovereign funds diversification

Built for OJK IKD approval from day one

OJK SANDBOX MANDATORY JAN 2025 • IKD DIGITAL INNOVATION TRACK

- **Mandatory sandbox since Jan 2025** — all digital assets must be evaluated before licensing. PALCapital files as IKD pilot.
- **Permissioned, KYB-only, consumer protection first** — ERC-3643 enforces at smart contract level, non-whitelisted wallets auto-revert.
- **Bank Indonesia rails** — Digital IDR + regulated USD stablecoins, ISO20022 RTGS, no unregulated crypto touchpoint.
- **Read-only auditor node** —Blockscout for full traceability. Escrow bank determination by Danantara IC + MoF + OJK, operated by Mandiri Custody under tripartite deed - meets OJK IKD consumer protection
- **IDDB precedent 2024** — first tokenization approved in OJK sandbox, 12-month testing model → full license path.

OJK IKD FLOW

GATED 12 MO

1 • Q4 26	2 • TESTING	3 • LICENSE
Sandbox registration	100 wallets live audit	Full OJK IKD approval

COMPLIANCE STACK

ENFORCED ON-CHAIN

- ✓ **ERC-3643 T-REX:** IdentityRegistry.isVerified() == false → revert
- ✓ **Privacy:** Classified sonar/hull/weapon specs never on-chain, ZK-SNARK only
- ✓ **Security:** BSSN sovereign cloud, ISO 27001, FIPS 140-2 L3 HSM

De-risked path: Mirrors IDDB sandbox success — regulator node from day 1, Digital IDR settlement, no retail exposure in Phase 1.

Strategic Value — Why PT PAL Wins

Zero dilution

100% equity & operational control retained - no withdrawToPAL(), no bank covenants

State ownership intact

Velocity

Capital in days, supplier payments automated via smart escrow. Yard throughput +40%.

7-14 days to funding

Sovereign leadership

World's first naval shipyard to tokenize defense assets on BSSN sovereign cloud, , 4 validators need 3 to finalize - Indonesia pioneer.

First-mover in defense RWA

Transparency

Real-time dashboard for Danantara/MoD/OJK, Mandiri independent escrow, ESG

OJK read-only node

ECOSYSTEM VALIDATED BY

PT PAL INDONESIA | DANANTARA | BKI SURVEYOR | NAVAL GROUP | KRAKATAU STEEL

Phased roadmap to $1B+ scale

Each phase OJK gated. No core DB refactor from 100 to 1M+ wallets — proxy pattern ERC-1967.

PHASE 1 • Q4 2026

$150M cap

Institutional Pilot — Scorpene Unit 1

- 100 whitelisted wallets • Danantara anchor
- Hyperledger Besu private • BSSN sovereign cloud
- BKI multi-sig manual — geotagged app

PHASE 2 • 2027

$300M

Hybrid Scale — Frigate + Sub Unit 2

- IoT as supporting evidence (crane load, CNC %, RFID) - weld quality remains BKI manual
- 10K investors • SAP S/4HANA + MQTT
- Chainlink DON 3-of-5 consensus, 6 validators tolerate 2

PHASE 3 • 2028+

$1B+

Sovereign Market — Full order book

- Retail IKD • secondary trading • 1M wallets
- Public hybrid L2 • Digital IDR + USD stables
- World's first naval STO exchange

OJK sandbox gate at each phase • Read-only auditor node for MoD / OJK • 99.95% uptime SLA

Besu IBFT 500 TPS

RPO 1min / RTO 15min

Risks & Mitigations

PRE-CLEARED BY BSSN & OJK IKD FRAMEWORK

Classified data leak HIGH

- ZK-SNARK redaction — prove $500M value, zero sonar/hull/specs on-chain
- BSSN-compliant sovereign cloud + AES-256-GCM, TLS 1.3

Smart contract exploit MEDIUM

- ERC-3643 battle-tested, Slither/Mythril 100% coverage, ERC-1967 proxy upgradeable
- Besu IBFT 4 validators tolerates 1 failure, RPO 1min RTO 15min

Supplier non-delivery MEDIUM

- Direct escrow to vendor only after BKI 2-of-2 multi-sig + IoT proof
- No advance to PAL ops account — zero diversion guarantee

Investor Distrust PAL Run Platform • MITIGATED

- 4 validators need 3 to finalize (PAL only 2)
- Mandiri Custodian owns fiat escrow under Tripartite Deed determined by Danantara+MoF+OJK, no withdrawToPAL() function exists
- code audited Slither/Mythril + formal verification, proxy upgrade 3-day timelock + 3-of-5 multisig (PAL+Danantara+BSSN) visible to OJK 72h, HSM Shamir 3-of-5 key split, OJK read-only node

Regulatory delay LOW • GATED

- OJK sandbox pre-consultation done, read-only auditor node for MoD/OJK from day 1
- Bank Indonesia rails: Digital IDR + regulated stablecoins, ISO20022

Investor concentration LOW • MITIGATED

- Danantara $900B anchor + state banks (Mandiri/BRI) + sovereign funds diversification, 100 whitelisted wallets Phase 1
- No retail exposure until Phase 3 — institutional-only IKD pilot

Result: All risk vectors covered by technical, legal, and financial controls. SRS verification matrix: ZK proof validates PO value with 0 specs revealed • Non-whitelisted transfer reverts • Escrow releases only after 2-of-2 signatures.

PHASE 1 PILOT READY — 45 DAYS TO GO-LIVE

Technology built. Regulatory path clear. Market ready.

1. **Board approval** for Phase 1 pilot ($150M PAL-SUB-1) as secured sovereign receivable
2. **OJK IKD sandbox registration** — joint PT PAL + Danantara with Mandiri as independent escrow agent (determined by Danantara IC + MoF + OJK)
3. **Mandate BKI** as independent validator & surveyor oracle — turnkey appliance provided by PAL, operated by BKI with own HSM key
4. **Deploy Besu network** on BSSN sovereign cloud — 2 validators on PAL on-prem Surabaya + Jakarta DC (BSSN compliant), 1 validator Danantara DC, 1 validator BKI - 4 nodes need 3 to finalize
5. **Launch Investor Portal + Treasury Dashboard + Surveyor App** with Secure Enclave 2-of-2 multi-sig - Go-live 90 days

By approving PALCapital, we don't just solve cash flow — **we make Indonesia the global standard for sovereign asset tokenization.**